

Corruption Robustness and Confidence Calibration in Image Classification using ResNet18 and CIFAR-10-C

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Abstract—Deep learning models have achieved remarkable success in image classification tasks when evaluated on clean benchmark datasets. However, their performance often deteriorates under common real-world image corruptions such as blur, noise, weather effects and compression artifacts. In addition to reduced accuracy, these models frequently produce overconfident predictions, making them unsuitable for safety-critical applications where prediction reliability is equally important as classification performance. This work investigates the robustness and confidence calibration of a ResNet18 image classification model using the CIFAR-10-C benchmark. Four representative corruption types, namely Gaussian Noise, Motion Blur, Fog and JPEG Compression, are analysed across multiple severity levels. Model calibration is evaluated using Expected Calibration Error (ECE), while robustness is measured through classification accuracy under corrupted conditions. Furthermore, three mitigation strategies including data augmentation, test-time normalization and selective prediction are investigated to improve model reliability. The study provides a practical evaluation of robustness challenges in deep learning and highlights the importance of confidence calibration for deploying image classification systems in real-world environments.

Index Terms—Image Classification, Deep Learning, Corruption Robustness, Confidence Calibration, ResNet18, CIFAR-10-C, Computer Vision

1. Introduction

Deep learning has significantly advanced the field of computer vision over the last decade. Convolutional Neural Networks (CNNs) have consistently achieved state-of-the-art performance on image classification benchmarks and are now widely used in applications such as autonomous driving, healthcare, surveillance and industrial inspection. Despite their excellent performance on clean datasets, these models often experience a substantial decline in accuracy when images contain distortions that commonly occur in real-world environments.

Real-world images are rarely captured under ideal conditions. Factors such as camera motion, environmental conditions, sensor limitations and image compression introduce

degradations including noise, blur, fog and compression artifacts. These corruptions create a distribution shift between training and testing data, reducing the generalization ability of deep learning models. Moreover, neural networks often remain highly confident even when making incorrect predictions, leading to poorly calibrated confidence estimates.

Model robustness and calibration have therefore become important research topics. Robustness measures how well a model maintains its predictive performance under input perturbations, while calibration evaluates whether prediction confidence accurately reflects the probability of correct classification. Both characteristics are essential for deploying machine learning systems in domains where reliability is critical.

This project investigates corruption robustness and confidence calibration using the CIFAR-10-C benchmark. A ResNet18 model is trained on the CIFAR-10 dataset and evaluated under multiple corruption types and severity levels. In addition to analysing model degradation, three mitigation techniques are explored: data augmentation, test-time normalization and selective prediction.

The main contributions of this work are summarized as follows:

- Evaluation of ResNet18 under multiple real-world image corruptions.
- Analysis of model performance across different corruption severity levels.
- Assessment of prediction reliability using Expected Calibration Error.
- Comparative study of three robustness improvement techniques.

2. Related Work

Although deep learning models have demonstrated remarkable classification accuracy on benchmark datasets, numerous studies have shown that their performance deteriorates significantly when evaluated on corrupted or distribution-shifted images. Hendrycks and Dietterich introduced the CIFAR-10-C benchmark to provide a standardized evaluation framework for corruption robustness across multiple corruption categories and severity levels. Their benchmark has become one of the most widely adopted datasets for evaluating robustness in image classification.

Confidence calibration has also received increasing attention in recent years. Guo et al. demonstrated that modern deep neural networks frequently produce overconfident predictions despite achieving high classification accuracy. They proposed temperature scaling as an effective post-processing calibration technique and highlighted the importance of confidence estimation in practical applications.

Several robustness enhancement techniques have been proposed in the literature. Data augmentation improves model generalization by exposing the network to a wider range of training samples. Test-time normalization attempts to reduce distribution mismatch during inference by normalizing corrupted inputs before prediction. Another practical approach is selective prediction, where the model rejects predictions with low confidence instead of forcing a potentially incorrect decision.

Motivated by these developments, this project evaluates robustness, calibration and mitigation techniques within a unified experimental framework using ResNet18 and the CIFAR-10-C benchmark.

3. Dataset

The experiments are conducted using the CIFAR-10 dataset together with the CIFAR-10-C corruption benchmark.

CIFAR-10 consists of 60,000 colour images with a spatial resolution of 32×32 pixels distributed across ten object categories including airplane, automobile, bird, cat, deer, dog, frog, horse, ship and truck. The dataset contains 50,000 training images and 10,000 testing images and has become one of the most commonly used benchmarks for image classification research.

To evaluate robustness under real-world image degradations, the CIFAR-10-C dataset is used. This benchmark extends the original CIFAR-10 test set by applying nineteen different corruption types at five severity levels. Instead of evaluating every corruption category, this project focuses on four representative distortions that frequently occur in practical computer vision applications.

- Gaussian Noise
- Motion Blur
- Fog
- JPEG Compression

Gaussian Noise simulates sensor noise introduced during image acquisition. Motion Blur represents distortions caused by camera or object movement. Fog models reduced visibility due to atmospheric conditions, while JPEG Compression introduces compression artifacts that commonly occur during image storage and transmission. Together, these corruption types provide a representative evaluation of robustness against diverse image degradations.

4. Methodology

The overall workflow of the proposed system consists of four major stages: model training, corruption robustness

evaluation, confidence calibration analysis and mitigation strategy comparison.

4.1. Model Architecture

A ResNet18 convolutional neural network is selected as the backbone classifier because of its balance between computational efficiency and classification performance. ResNet18 employs residual learning through skip connections, allowing deeper architectures to be optimized effectively while reducing vanishing gradient problems. The final fully connected layer is configured to classify the ten CIFAR-10 object categories.

4.2. Training Procedure

The model is trained using the clean CIFAR-10 training dataset. Prior to training, all images are resized where necessary, converted into tensors and normalized using the dataset mean and standard deviation. Cross-entropy loss is used as the optimization objective, while the Adam optimizer updates model parameters during training. Training continues for multiple epochs until the model converges on the training data.

4.3. Corruption Robustness Evaluation

Following training, the model is evaluated on corrupted images from the CIFAR-10-C benchmark. Four corruption categories are considered, with each corruption evaluated across five severity levels. Classification accuracy is measured for every corruption-severity combination to analyse the degradation in predictive performance as image quality decreases.

4.4. Confidence Calibration

In addition to classification accuracy, confidence calibration is evaluated using Expected Calibration Error (ECE). This metric measures the difference between prediction confidence and observed accuracy across confidence intervals. A lower ECE indicates better alignment between confidence scores and prediction correctness, resulting in a more trustworthy classification model.

4.5. Mitigation Strategies

Three complementary approaches are investigated to improve robustness and reliability.

4.5.1. Data Augmentation. Data augmentation increases training diversity by applying random image transformations such as horizontal flipping, random cropping and normalization. These transformations encourage the model to learn invariant feature representations and improve generalization to unseen corruptions.

4.5.2. Test-Time Normalization. Test-time normalization standardizes corrupted input images before inference. By reducing distribution mismatch between training and testing data, the model becomes more resilient to moderate corruption levels without requiring retraining.

4.5.3. Selective Prediction. Selective prediction introduces a confidence threshold during inference. Predictions with confidence below the threshold are rejected rather than accepted automatically. Although this reduces prediction coverage, it significantly improves the reliability of accepted predictions and is particularly useful in safety-critical environments.

The proposed methodology provides a comprehensive framework for analysing both robustness and calibration while comparing practical mitigation techniques for improving the reliability of deep learning image classifiers.

5. Experimental Setup

The experiments were conducted using the PyTorch deep learning framework. A ResNet18 convolutional neural network was selected because it provides a good balance between computational efficiency and classification performance. The network was trained using the CIFAR-10 training dataset and evaluated on both the clean CIFAR-10 test set and the corrupted CIFAR-10-C benchmark.

Images were normalized using the standard CIFAR-10 mean and standard deviation before being passed to the network. Cross-entropy loss was used as the optimization objective, while the Adam optimizer was employed to update model parameters during training.

To evaluate robustness, four representative corruption categories were selected from CIFAR-10-C:

- Gaussian Noise
- Motion Blur
- Fog
- JPEG Compression

Each corruption was evaluated across five severity levels. Classification accuracy was used as the primary robustness metric, while Expected Calibration Error (ECE) was used to analyse confidence calibration.

Three mitigation techniques were implemented and compared.

- 1) Data Augmentation
- 2) Test-Time Normalization
- 3) Selective Prediction

These methods were evaluated to determine their ability to improve prediction robustness and confidence calibration under corrupted image conditions.

6. Results and Discussion

The trained ResNet18 model achieved strong classification performance on clean CIFAR-10 images. However, introducing image corruptions resulted in a gradual degradation in prediction accuracy. The extent of degradation depended on both the corruption type and its severity.

6.1. Performance on Clean Images

The baseline evaluation on the original CIFAR-10 test set demonstrated that the trained network successfully learned discriminative visual features for all ten object classes. This baseline serves as the reference point for evaluating robustness under corrupted inputs.

6.2. Performance under Image Corruptions

The corruption experiments reveal that model performance decreases consistently as corruption severity increases. Mild corruptions introduce only a small reduction in classification accuracy, whereas severe corruptions substantially alter image characteristics, making object recognition considerably more difficult.

Among the evaluated corruptions, Gaussian Noise introduces random pixel variations that obscure important visual patterns. Motion Blur removes edge information and fine details required for accurate recognition. Fog reduces image contrast and visibility, while JPEG Compression introduces block artifacts that distort texture information.

Overall, the experimental observations demonstrate that the model is sensitive to distribution shifts introduced by common real-world image degradations.

6.3. Severity Analysis

The evaluation demonstrates that classification performance decreases as corruption severity increases. Lower severity levels preserve most visual information and therefore have only a limited impact on model predictions. However, increasing corruption intensity progressively removes discriminative image features, resulting in lower prediction confidence and higher classification errors. This behaviour confirms that common image corruptions introduce distribution shifts that significantly affect deep learning model performance.

6.4. Confidence Calibration

Besides the reduction in classification accuracy, corrupted images also influence prediction confidence. The model frequently assigns high confidence scores to incorrect predictions, indicating calibration errors under distribution shift. Expected Calibration Error (ECE) was therefore used to evaluate the agreement between predicted confidence and observed accuracy. Better calibration corresponds to smaller confidence errors and more reliable predictions.

6.5. Evaluation of Mitigation Techniques

Three mitigation strategies were investigated to improve robustness and prediction reliability. Data augmentation improved model generalization by exposing the network to a wider range of image variations during training. Test-time normalization reduced the mismatch between training

and testing distributions, while selective prediction increased prediction reliability by rejecting uncertain classifications. Overall, these approaches demonstrated their usefulness in improving robustness against common image corruptions while enhancing the trustworthiness of model predictions.

7. Future Work

Future research can extend this work by investigating larger image datasets such as ImageNet-C and evaluating more recent deep learning architectures. Advanced robustness techniques including adversarial training, self-supervised learning and uncertainty estimation methods may further improve prediction reliability.

Additional calibration techniques such as temperature scaling, ensemble learning and Bayesian neural networks can also be explored for safety-critical applications.

8. Conclusion

This work presented a comprehensive evaluation of corruption robustness and confidence calibration for image classification using a ResNet18 model trained on CIFAR-10 and evaluated on the CIFAR-10-C benchmark.

Experimental analysis demonstrated that model performance decreases progressively as corruption severity increases. In addition to reduced classification accuracy, corrupted inputs resulted in poorer confidence calibration, highlighting the importance of evaluating both robustness and prediction reliability.

Among the investigated mitigation strategies, data augmentation improved generalization while selective prediction enhanced prediction reliability by reducing overconfident incorrect classifications. Overall, the study demonstrates that robustness evaluation and confidence calibration should be considered together when deploying deep learning models in real-world computer vision applications.

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